

Problem sheet 4: Quantum Channels

Problem 1 (HW). Writing down channels. For the following scenarios, write down the quantum channel that models them.

- (a) We have a qubit system. With probability $1/2$ we do nothing and with probability $1/2$ we discard the state and prepare the state $|0\rangle$.
- (b) We have two qubit systems A and B . We replace the state on the B -system with $|0\rangle$ and apply a Pauli Z operator to the A -system.
- (c) We start with system A , add system E in state $|0\rangle$. We apply a global unitary U on A and E and then discard the system E .

Problem 2 (HW). More examples of channels. Prove that the superoperators defined at lecture are in fact quantum channels.

- (a) Depolarizing channel. *Hint: compute the Choi operator.*
- (b) Dephasing channel. *Hint: find a Kraus representation.*
- (c) Erasure channel.
- (d) Replacement channel. *Hint: Prepare a purification of ρ and discard both the input system A and the purifying system. This gives a Stinespring representation of the replacement channel.*

Problem 3. Even more examples of channels.

- (a) Consider the superoperator $\Phi_{A \rightarrow AB}$ defined by

$$M_A \mapsto M_A \otimes \rho_B,$$

for some fixed $\rho \in S(B)$. Show that this defines a quantum channel and compute its Choi operator.

- (b) Consider a measurement $\mu_A = \{\mu_A(x)\}_{x \in \Omega}$ and states $\rho_{B,x} \in S(B)$, for $x \in \Omega$. Define a superoperator $\Phi_{A \rightarrow B}$ by

$$\Phi_{A \rightarrow B}[M_A] = \sum_{x \in \Omega} \text{tr}[\mu_A(x) M_A] \rho_{B,x}.$$

Show that this defines a quantum channel and give Kraus representation.

- (c) Let $U_1, \dots, U_r \in \text{Lin}(A)$ be unitaries and let p_i , $i = 1, \dots, r$ be a probability distribution. Let Φ_A be the superoperator defined by

$$\Phi_{A \rightarrow B}[M_A] = \sum_{i=1}^r p_i U_i M_A U_i^\dagger$$

Show that this defines a quantum channel and give a Stinespring representation

- (d) For each of the above channels, give an operational interpretation.

Problem 4. Constructions for completely positive maps. This provides an alternative proof of the characterization of CP maps.

- (a) Suppose we are given $\Phi_{A \rightarrow B}: \text{Lin}(A) \rightarrow \text{Lin}(B)$ in its Stinespring representation

$$\Phi_{A \rightarrow B}[M_A] = \text{tr}_E[V M_A V^\dagger],$$

for $V \in \text{Lin}(A, BE)$. Use V to directly construct Kraus operators $X_i \in \text{Lin}(A, B)$ for $i = 1, \dots, r$ such that

$$\Phi_{A \rightarrow B} = \sum_{i=1}^r X_i M_A X_i^\dagger.$$

- (b) Suppose that we are given $\Phi_{A \rightarrow B}$ in its Kraus representation as above. Write J_{AB}^Φ in terms of the X_i , and show that it is positive.
- (c) Assuming only that J_{AB}^Φ is positive, show that $\Phi_{A \rightarrow B}$ is completely positive. That is, show that for any auxiliary system R and $M_{AR} \in \text{Lin}(AR)$,

$$M_{AR} \geq 0 \implies (\Phi_{A \rightarrow B} \otimes \mathcal{I}_R)[M_{AR}] \geq 0.$$

Problem 5. Uniqueness of representations. Let $\Phi_{A \rightarrow B}$ be a quantum channel with two different Stinespring representations

$$\Phi_{A \rightarrow B} = \text{tr}_E[V M_A V^\dagger] = \text{tr}_F[W M_A W^\dagger],$$

for $V \in \text{Isom}(A, BE)$ and $W \in \text{Isom}(A, BF)$ isometries. Let $\mathcal{V}_{A \rightarrow BE}$ and $\mathcal{W}_{A \rightarrow BF}$ be the quantum channels applying the isometries V and W , respectively.

- (a) Show that the Choi states of the isometric channels $\mathcal{V}_{A \rightarrow BE}$ and $\mathcal{W}_{A \rightarrow BF}$ are purifications of the Choi state of $\Phi_{A \rightarrow B}$.
- (b) Suppose that $\dim(\mathcal{H}_E) > \dim(\mathcal{H}_F)$. Show that $V = (\mathbb{1}_B \otimes U)W$, where U is an isometry mapping F to E .
- (c) Suppose that we have two different Kraus representations of $\Phi_{A \rightarrow B}$:

$$\Phi_{A \rightarrow B}[M_A] = \sum_i^r X_i M_A X_i^\dagger = \sum_i^s Y_i M_A Y_i^\dagger.$$

Suppose that $r \geq s$. Show that there are $u_{ij} \in \mathbb{C}$ such that

$$X_i = \sum_{j=1}^s u_{ij} Y_j,$$

where the u_{ij} are from an isometric matrix. *Hint: construct Stinespring extensions based on the different Kraus representations.*